

Data-Driven Pitch Optimization Progress Report (Team 46)

Team Project - Georgia Tech M.S. in Analytics Program

I. INTRODUCTION

We are creating a tool that classifies baseball pitches by speed, location, spin rate, and movement to identify the areas of the strike zone where hitters struggle the most. The tool takes raw pitch data and uses a clustering algorithm to transform it into easy-to-interpret, actionable insights and then presents the output on an interactive dashboard. It is intended for coaches, players, and analysts to make smarter, data-based decisions during preparation for games.

II. PROJECT PROBLEM DEFINITION

Pitch selection and batter scouting rely on primitive scouting, video analysis, and overall analytics that mostly provide broad, league-level trends. These resources do not combine significant pitch statistics such as velocity, spin rate, and location to provide individualized insights. Our project fills that void by grouping similar pitches to find specific areas where batters have low batting averages. The interactive dashboard will enable users to sift through results by player or pitcher, uncover hidden trends, and obtain recommendations on the best pitch type and location for given situations. This personalized mechanism will optimize pitch selection and overall game performance.

III. LITERATURE SURVEY

We conducted an extensive review of 18 sources to ground our approach. Notable works include:

- **Albert (2010): Using the Count to Measure Pitching Performance** - Explores how pitch count affects performance and decision-making. Adds situational context to pitch analysis. Presents findings statistically without a visual representation [1].
- **Dvorocsik et al. (2020): Using Clustering to Find Pitch Subtypes and Effective Pairings** - Uses clustering techniques to identify pitch subtypes. Provides a basis for multi-dimensional clustering. Lacks direct linkage to hitter-specific vulnerabilities [2].
- **Hamel (2021): Quantifying Pitch Tunneling: The Role of Deception in Pitch Sequencing** - Explores how similar initial pitch trajectories can diverge later to deceive hitters. Informs strategies for effective pitch sequencing and pairing. Does not directly translate to recommendations for optimal pitch selection [3].
- **Healey (2015): Modeling the Probability of a Strikeout for a Batter/Pitcher Matchup** - Enhances strikeout prediction models using matchup data. Provides a framework for incorporating player-specific statistics. Does not leverage pitch-level details for optimizing recommendations [4].
- **Kennedy-Shaffer (2024): The Effects of Major League Baseball's Ban on Infield Shifts: A QuasiExperimental Analysis** - Evaluates batter performance changes post-shift ban. Offers context on pitch effectiveness under varying conditions. Focuses on league-wide trends rather than individualized pitch selection [5].
- **L'Oiseau (2019): Investigating the Importance of Pitch Selection with Clustering** - Uses clustering to analyze pitch selection and its impact on game outcomes. Directly relates to our clustering methodology. Does not explore variation in clusters across the strike zone [6].
- **Marshall (2016): Reverse Engineering Swing Mechanics from Statcast Data** - Analyzes swing mechanics using Statcast data to reveal hitter behavior. Provides insights into hitter weaknesses. Requires additional data to link swing mechanics with optimal pitch selection [7].
- **Martin (2019): Predicting MLB Strikeout Rates from Differences in Velocity and Movement among Player Pitch Types** - Examines how variations in pitch velocity and movement affect strikeouts. Identifies key variables for effective clustering. Omits detailed analysis of pitcher/batter matchup dynamics [8].
- **Pavitt (2024): Plummeting Batting Averages Are Due to Far More than Infield Shifting: Part Two: Strikeouts** - Links declining batting averages to strikeout rates and pitch characteristics. Provides league-wide trends in pitch attributes. Lacks granularity at the individual matchup level [10].
- **Pomrenke (2020): Using Clustering to Find Pitch Subtypes and Effective Pairings** - Identifies effective pitch pairings through clustering. Supports our goal of finding optimal pitch combinations. Lacks tailored recommendations based on individual matchups [11].
- **Side & Tran (2018a): Using Multi-Class Classification Methods to Predict Baseball Pitch Types** - Uses statistical models to predict the next pitch based on various game factors. Highlights the role of situational variables in pitch prediction. Focuses on aiding hitters rather than optimizing pitcher decisions [12].
- **Side & Tran (2018b): Classifying Pitch Types in Baseball Using Machine Learning Algorithms** - Tests multiple classification models to improve pitch classifica-

tion accuracy. Demonstrates the potential for integrating situational factors into pitch classification. Does not utilize clustering techniques [13].

- **Spangler (2017): An Exploration of the First Pitch in Baseball** - Investigates the strategic significance of the first pitch using logistic regression. Underlines the importance of the initial pitch in establishing an advantage. Does not propose methods to enhance first-pitch outcomes [14].
- **USPROC (2018): Fixing the Curve** - Utilizes model-based clustering for pitch classification. Identifies key metrics for grouping pitches. Focuses on classification rather than addressing hitter weaknesses [16].
- **Ursin (2014): A Markov Model for Baseball** - Applies Markov models to simulate pitch sequences. Offers perspective on pitch sequencing. Does not incorporate unsupervised learning for matchup-specific insights [15].
- **Wilson (2019): Pitch Quantification Part 1** - Introduces the Quality of Pitch (QOP) metric, incorporating speed, location, and movement. Suggests standardized measures for pitch quality. Does not account for individual hitter weaknesses or pitch location analysis [17].
- **Zhao (2015): Pitch-to-Pitch Correlation** - Studies correlations among pitch attributes and their role in predicting strikeout rates. Emphasizes the importance of pitch sequencing. Does not offer personalized matchup recommendations [18].
- **Otembra (2022): SmartPitch: Applied Machine Learning for Professional Baseball Pitching Strategy** - Analyzes pitcher and batter characteristics using neural networks. Uses Markov Decision Processes to model the count for any matchup and proposes a reward maximizing pitching strategy using reinforcement learning [9].

These studies collectively underscore the need for a more individualized, matchup specific analysis, which is the exactly the gap our project aims to address.

IV. PROPOSED METHOD

In the “Batters” script, we joined datasets containing player names, at bat result data, and pitch-by-pitch data. We then reduced the dataset to only include rows for the final pitch of each at bat because for this dataset we were focused only on the result of the at bat. We removed NaN values and combined the number of balls and strikes into one column to get the count of the at bat when the result occurred. Selecting the desired columns, we ended up with a dataset containing the player name, the handedness of the pitcher (righty or lefty), the count of the at bat, the pitch type that was thrown, and the result of the at bat. From this dataset, we created five additional datasets: grouping by player, grouping by player and pitcher handedness, grouping by player and count, grouping by player and pitch type, and overall league data for each at bat result. Each of these datasets shows the number of times each at bat result occurred.

This script and the resulting datasets were created for additional analysis in Tableau that could be used as sup-

plemental information on our dashboard. While the primary purpose of the dashboard is to provide pitch recommendations, these datasets could be used to calculate statistics like batting average, on base percentage, and slugging percentage for a selected batter. The statistics could also be shown for different scenarios, such as a specific at bat count, pitcher handedness, or pitch type thrown. We feel that this will provide additional insights to supplement our pitch recommendations.

V. INNOVATIONS AND VISUALIZATIONS

A. Clustering and Filtering

In the clustering script, players with fewer than 1,000 at-bats over the three-year dataset were filtered out to ensure sufficient data for meaningful results. Additionally, pitches with extreme values were removed to exclude those thrown far outside the strike zone. While these pitches may have ended an at bat, batters are far less likely to swing at them, making their inclusion less relevant to the analysis.

Once the player dataset is finalized, multiple clustering models will be tested with varying numbers of clusters to determine the optimal grouping. Separate clustering models will be applied to pitches thrown by left-handed and right-handed pitchers. A Python function will then identify the optimal cluster assignment for each player.

B. Tableau Integration and Additional Visuals

After each pitch is assigned to a cluster, the results are imported into Tableau. Tableau calculates the batting average for each cluster and recommends the three clusters with the lowest batting averages. These recommendations include the ideal pitch location and pitch type. If a cluster spans multiple zones, the most frequently targeted zone and pitch type will be highlighted as the optimal choice. In Tableau, when a specific pitcher is selected, the dataset is filtered to identify the optimal pitches to throw based on the chosen player. This analysis also considers the pitcher’s maximum velocity and spin rates to ensure recommendations align with their capabilities.

Future innovations and in-progress visuals include:

- **Batter selection and strikezone heatmap (Player Scorecard):** By selecting an individual batter, you are able to see his batting average, slugging percentage, and strikeout rate along with a heatmap of where the hitter is likely to succeed. This can be paired with filters such as pitcher throwing hand, exact pitcher, and specific pitch to help both pitcher and hitter identify weaknesses.
- **Batter vs. Pitcher visual:** Looks at an individual pitcher to show the percentage of pitches that pitcher usually throws. It also displays, across the entire league, how often that pitch is thrown by other pitchers. The visual compares individual versus league percentages, showing if a pitcher has a tendency to pitch more fastballs (for example) than the league average. For the batter, it shows the likely outcome of the at bat by the individual batter. Clicking on a pitch from the pitcher will reveal the likely outcome that the batter will face.

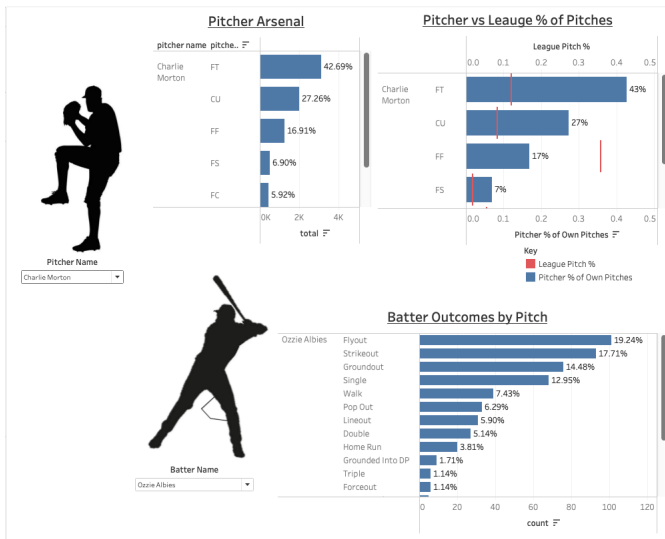


Fig. 1. Batter vs Pitcher visual

- **Pitch Types Script:** Joins data from three datasets to provide a summary of pitch type and frequency for each pitcher as well as overall league-wide numbers, which may be utilized in determining the best pitch for a particular pitcher-batter matchup.

VI. DESCRIPTION OF UPCOMING EXPERIMENTS

List of questions:

- How well do the clustering algorithm's pitch recommendations perform versus real-world pitcher performance?
- Does it perform equally across all at-bat scenarios and pitcher-batter matchups?
- Does the clustering algorithm generalize to new data?
- Is the feature set appropriate?

To validate the clustering algorithm results and overall methodology, the pitch recommendations can be evaluated against new pitch data from MLB games. In near real-time we can query new pitches and at-bats using Baseball Savant's gamefeed. Alternatively, the pybaseball library in Python offers the same data in batches after games are complete. The experiments will focus on comparing the clustering algorithm recommendations with the actual result of the pitch. The additional statistics mentioned before, such as offering insights for different at-bat counts, can also be tested this way to ensure accuracy is consistent across all levels of detail including, for example, the more critical pitches where a full count occurs.

To ensure that the algorithm's clusters are not overly complex, another experiment could take subsets from historical pitch data to ensure that the recommendations align with real performance trends. In a similar way this method would involve comparing the pitch suggestions with historical pitch results in different at-bat scenarios. This could also include removing or adding variables such as exit velocity and launch angle to refine the pitch suggestions. After rerunning the clusters and refreshing the dashboard, the new recommendations

could be compared again with new pitch data to test the effectiveness of that feature set.

An extension of the experiments would be to explore batter and pitcher adjustments over a season. Future iterations could involve predictive modeling to detect patterns or trends of a certain pitch. For pitchers, a fastball might be working at the beginning of a season, but lose its effectiveness later in the season as the pitcher may experience fatigue. Conversely, this might improve a batter's performance of that same pitch, tilting the scales of the fastball. Using predictive modeling to understand these relationships could bring more depth to the clustering analysis by making adjustments to pitch recommendations throughout the season based on trends.

VII. PLAN OF ACTIVITIES

Completed Tasks:

- Data collection and initial cleaning.
- Development of the initial clustering script.
- Integration of the "Batters" script and generation of supplemental datasets.

Ongoing Tasks:

- Refinement of clustering models for both left-handed and right-handed pitchers.
- Importing and visualization of cluster results in Tableau.
- Development of additional visuals, including the Player Scorecard and Batter vs. Pitcher comparisons.

Upcoming Tasks:

- Conduct experiments to validate the clustering recommendations.
- Expand and adjust the feature set based on experimental outcomes.
- Finalize the integration of all components and complete documentation.

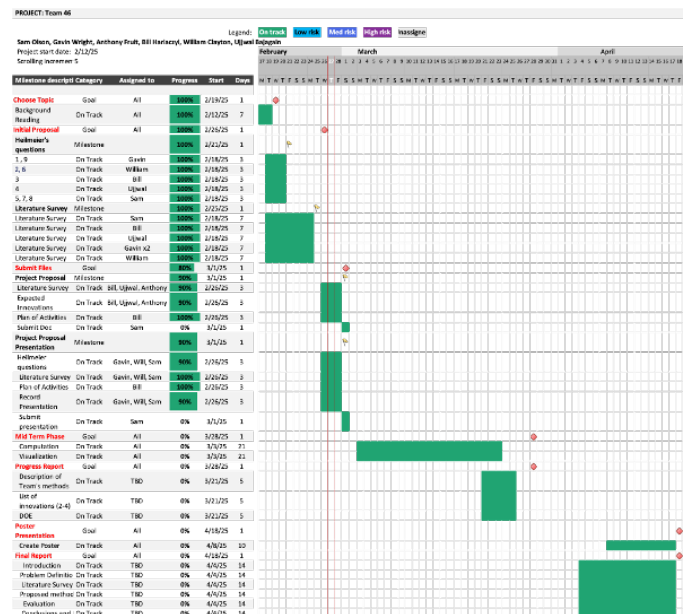


Fig. 2. Gantt Chart from Proposal Document



Fig. 3. Updated Gantt

VIII. CONCLUSION AND NEXT STEPS

Our project has made significant strides in transforming raw pitch data into actionable insights through a combination of data cleaning, advanced clustering, and interactive visualizations. We have established a robust data pipeline and developed initial scripts that integrate pitch-by-pitch information into a format suitable for in-depth analysis. The extensive literature survey underscores the relevance and necessity of our approach, while our innovative methods promise a tailored, matchup-specific analysis that traditional scouting and analytics fail to deliver.

Looking ahead, our next steps are clear: we will refine our clustering algorithms, validate our recommendations with new and historical pitch data, and enhance our Tableau dashboard with advanced visualizations. Experimental validation and feature-set optimization remain key priorities, ensuring that our tool not only performs well in controlled analyses but also translates effectively to real-world game scenarios.

Lastly, our research hopes to bridge the gap between pitch decision-making and statistical inquiry, providing coaches, players, and analysts with a useful instrument with which to improve pitch decisions.

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